Physics 10.1.23

Electric charge & potential

Electric charge: Charge is the fundamental and characteristics property of elementary particle.

Elementary particles

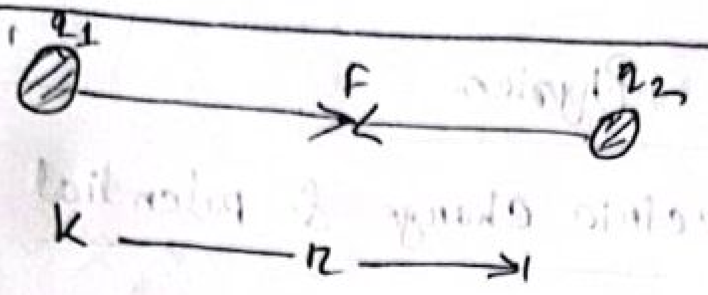
(i) Electron

(ii) Proton

(iii) Neutron.

Value of charge is 1.6×10^{-19} C.

Coulomb's Law: The force of attraction or ~~refulction~~ repulsion between two charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.



$F_e \propto \frac{q_1 q_2}{r^2}$ $\qquad$ $F_e \propto q_1 q_2$ —— (i)

$F_e = c \cdot \frac{q_1 q_2}{r^2}$ $\qquad$ $F_e \propto \frac{1}{r^2}$ —— (ii)

$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_1 q_2}{r^2}$

$= 9 \times 10^9 \frac{q_1 q_2}{r^2}$

(Q) Show that which force is stronger?

⇒ Electrical force is stronger. [৫ [illegible] -তাই]

→ Prove?

Or, Show that electrical force is very much stronger than the gravictional force!

Solution:

From Coloumb's Law, we can write:

$$F_e = C \cdot \frac{q_1 q_2}{r^2} \quad \text{——— (i)}$$

Here,

$q_1 = 1C$, and $r = 1m$.
$q_2 = 1C$.

$$\Rightarrow F_e = 9 \times 10^9 \times \frac{1 \times 1}{1^2} \, Nm^{-2}C^2$$

$$= 9 \times 10^9 \, Nm^{-2}C^2$$

Again,

From Newton's law of graviational force we can write:

$$F_g = G \frac{m_1 m_2}{r^2}$$

$$= 6.67\cancel{8} \times 10^{-11} \cdot \frac{1 \times 1}{1^2}$$

$m_1 = 1\,kg$
$m_2 = 1\,kg$
$r = 1m$

$$= 6.67\cancel{8} \times 10^{-11}$$

$\therefore F_e > F_g$.

$\therefore$ So, F_e is very much stronger than F_g.

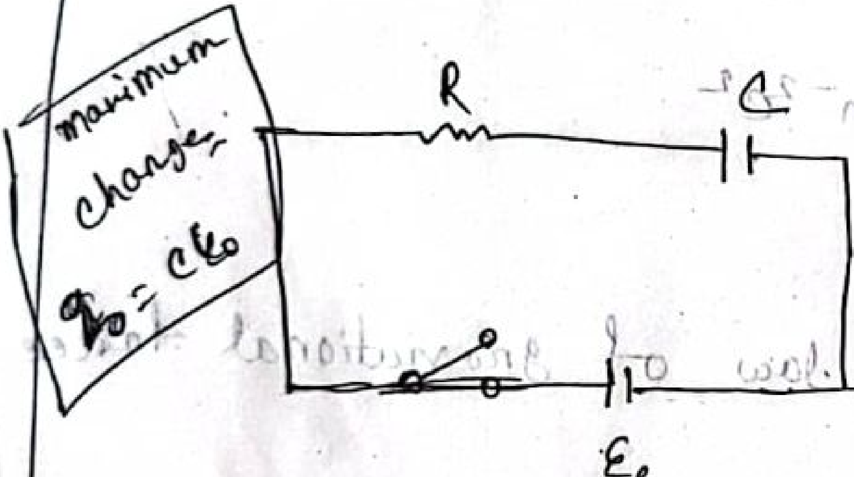
—o—

(1) Growth of a charge in capacitor / charging

(2) ~~Decay~~ Decay of a " " " / Discharging.

(1) Growth of a charge and Decay of a current

$$I = \frac{dQ}{dt}$$



[* Resistor → across → voltage, $V = IR$
* Capacitor " " ", $V = \frac{Q}{C}$]

The potential drop of across the resistor,

$$V = RI$$

The potential drop of across the capacitor,

$$V = \frac{Q}{C}$$

According to KVL we can write.

$$\mathcal{E}_0 = RI + \frac{q}{C}$$

$$\Rightarrow \mathcal{E}_0 = R\frac{dq}{dt} + \frac{q}{C}$$

$$\Rightarrow \mathcal{E}_0 - \frac{q}{C} = R.\frac{dq}{dt}$$

$$\Rightarrow \frac{C\mathcal{E}_0 - q}{C} = R.\frac{dq}{dt}$$

$$\Rightarrow C\mathcal{E}_0 - q = RC.\frac{dq}{dt}$$

$$\Rightarrow \frac{dq}{C\mathcal{E}_0 - q} = \frac{dt}{RC}$$

$$\Rightarrow \frac{dq}{q - C\mathcal{E}_0} = -\frac{dt}{RC}$$

Integrating Both sides,

$$\int \frac{dq}{q - C\mathcal{E}_0} = \int -\frac{dt}{RC}$$

$$\Rightarrow \ln(q - C\mathcal{E}_0) = -\frac{t}{RC} + K \quad \text{--- (1)}$$

When $t = 0$, $q = ?$

$$\ln(q - C\varepsilon_0) = -\frac{t}{RC} + k$$

$$\Rightarrow \ln(0 - C\varepsilon_0) = -\frac{0}{RC} + k$$

$$\Rightarrow \ln(-C\varepsilon_0) = k$$

$$\therefore k = \ln(-C\varepsilon_0)$$

~~k का मान~~

from eqn. (1)

$$\ln(q - C\varepsilon_0) = -\frac{t}{RC} + k$$

$$\Rightarrow \ln(q - C\varepsilon_0) = -\frac{t}{RC} + \ln(-C\varepsilon_0)$$

$$\Rightarrow \ln(q - C\varepsilon_0) - \ln(-C\varepsilon_0) = -\frac{t}{RC}$$

$$\cancel{\Rightarrow \ln q - \ln C\varepsilon_0 + \ln C\varepsilon_0 = -\frac{t}{RC}}$$

$$\Rightarrow \ln \frac{q - \varepsilon_0 C}{-\varepsilon_0 C} = -t/RC$$

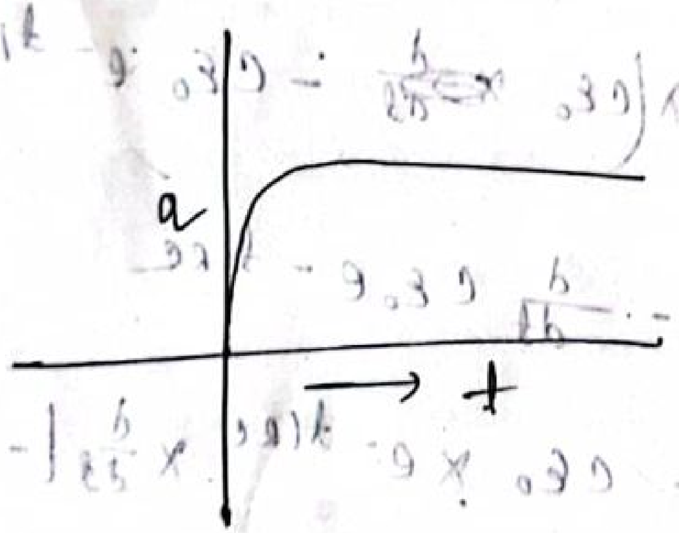
$\Rightarrow \dfrac{q - \varepsilon_0 C}{-\varepsilon_0 C} = e^{-t/RC}$

$\Rightarrow q - \varepsilon_0 C = -\varepsilon_0 C\, (e^{-t/RC})$

$\Rightarrow \cancel{q - \varepsilon_0 C} =$

$\Rightarrow q = C\varepsilon_0 - C\varepsilon_0 e^{-t/RC}$

$\Rightarrow q = C\varepsilon_0 (1 - e^{-t/RC})$



Decay of a current,

$$I = \frac{dQ}{dt}$$

$$= \frac{d}{dt}\left\{C\varepsilon_0\left(1 - e^{-t/RC}\right)\right\}$$

$$\cancel{= \frac{d}{dt} \times C\varepsilon_0 \times \frac{d}{dt} \times \left(1 - e^{-t/RC}\right)}$$

$$\cancel{= 0 \times 0 - e^{-t/RC} \times \frac{d}{dt}\left(-t/RC\right)}$$

$$= \frac{d}{dt} \times \left(C\varepsilon_0 - C\varepsilon_0 e^{-t/RC}\right)$$

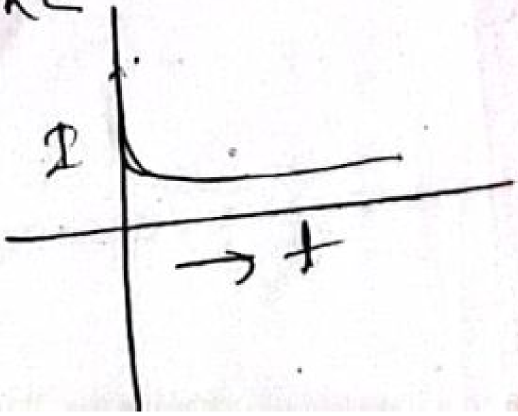
$$= \frac{d}{dt} C\varepsilon_0 - \frac{d}{dt} C\varepsilon_0 e^{-t/RC}$$

$$= 0 - C\varepsilon_0 \times e^{-t/RC} \times \frac{d}{dt}\left(-\frac{t}{RC}\right)$$

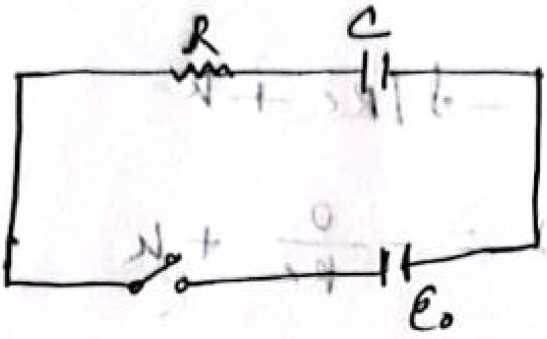
$$= -C\varepsilon_0 \times e^{-t/RC} \times \frac{-1}{RC}$$

$$= \frac{C\varepsilon_0}{RC} \times e^{-\frac{t}{RC}}$$

$$= \frac{\varepsilon_0}{R} e^{-t/RC}$$



(a) Decay of charge in a capacitor and growth of a current.



When $E_0 = 0$, q = max.

So, we can write, $q = q_0 = CE_0$

The potential drop of across the resistor,

$$V = RI$$

The potential drop of across the capacitor, $V = \frac{q}{C}$.

$\therefore \cancel{V = \frac{q}{C}}$

$\Rightarrow \cancel{\frac{q}{CE_0}}$

From eqn. (i).

$$\ln(q - C\varepsilon_0) = -t/RC + K$$

$$\cancel{\Rightarrow \ln(C\varepsilon_0 - C\varepsilon_0) = -t/RC + K}$$

$$\Rightarrow \ln(q_0 - C\varepsilon_0) = -\frac{0}{RC} + K$$

$$\Rightarrow \ln(C\varepsilon_0 - C\varepsilon_0) = K$$

$$\Rightarrow K = q$$

~~putting the value on eqn (i).~~

$$\cancel{\ln(q - C\varepsilon_0)} =.$$

Eq (i) $\Rightarrow$.

$$\ln(q - C\varepsilon_0) =$$

$$\mathcal{E}_0 = IR + \frac{q}{c} = 0$$

$$\Rightarrow \frac{q}{c} = -iR$$

$$\Rightarrow \frac{q}{c} = -\frac{dq}{dt} R$$

$$\Rightarrow \frac{q}{c} = \frac{dq}{dt}$$

$$\Rightarrow \frac{dq}{q} = -\frac{dt}{Rc}$$

Integrating

$$\int \frac{dq}{q} = \int -\frac{dt}{Rc}$$

$$\Rightarrow \ln q = -\frac{t}{Rc} + k \quad \text{—— (i)}$$

$$q = q_0 \quad @$$

$$\text{at } t = 0$$

$$\ln q_0 = k \quad \text{—— (ii)}$$

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putting the value of k in (i)

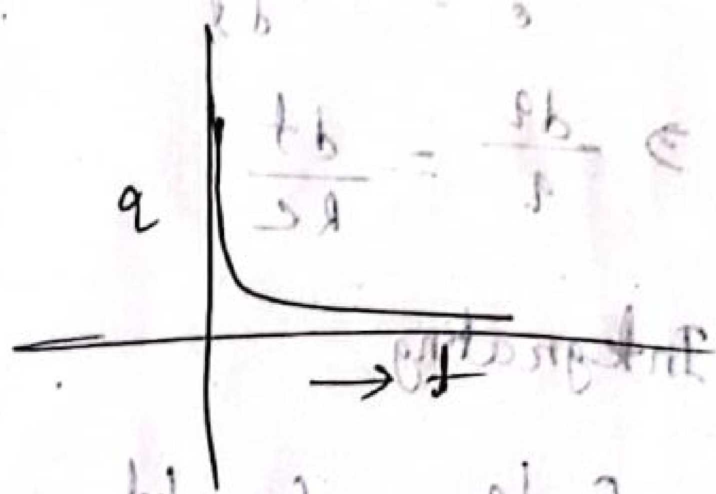
$$\ln q = -\frac{t}{RC} + \ln q_0$$

$$\Rightarrow \ln q - \ln q_0 = -\frac{t}{RC}$$

$$\Rightarrow \frac{q}{q_0} = e^{-t/RC}$$

$$\Rightarrow q = q_0 \cdot e^{-t/RC}$$

$$= C\varepsilon_0 \cdot e^{-t/RC}$$



Now,

$$I = \frac{dq}{dt}$$

$$= \frac{d}{dt}\left[C\varepsilon_0 \cdot e^{-t/RC}\right]$$

$$= -C\varepsilon_0 \frac{1}{RC}\left(e^{-t/RC}\right)$$

$$= -\frac{\varepsilon_0}{R} e^{-t/RC}$$

Problem 1: A 150f capacitor is connected through a 500 Ω resistor (40 V) battery.

i) What is the time constant of the circuit.

ii) What is the (final charge) q_0 on a capacitor plate.

iii) How long does it take for the charge on a capacitor ~~plate~~ take. ~~to~~ reach on a 0.8 q_0

i)

$$t = RC$$

$$= 500 \times 150$$

$$= 75,000\ s$$

ii) $q_0 = C\,\varepsilon_0$

$$= 150 \times 40$$

$$= 6000\ C.$$

33(i)

$$q = q_0(1 - e^{-t/RC})$$

$$\Rightarrow 0.8 = (1 - e^{-t/RC})$$

$$\Rightarrow e^{-t/RC} = -0.2$$

$$\Rightarrow -\frac{t}{RC} = -1.609$$

$$\Rightarrow t = RC \times (1.609)$$

$$\Rightarrow \quad = 75000 \times 1.609$$

$$= 120:675$$

A 50 μF capaciton anchanged in connected rough a 200 Ω nesiston to a 12 V battery.

What is the magnitude of the final change to on the capaciton.

What is the time constant of circuit?

) How long after the capacitor is connected to the battery will it be changed to 1/2 q_0 and to 0.90 q_0?

a)

$q = C\varepsilon_0$

$= 50 \mu F \times 12 V$

$= 6 \times 10^{-4} C$ $[50 \mu F = 5 \times 10^{-6} F]$

(b) $\tau = RC$

$= 300\Omega \times 50 \times 10^{-6}$

$= 0.015\,s$

(c) $q = q_0 (1 - e^{-t/RC})$

$\Rightarrow \frac{1}{2} q_0 = q_0 (1 - e^{-t/RC})$

$\Rightarrow \frac{1}{2} = 1 - e^{-t/RC}$

$\Rightarrow e^{-t/RC} = 1 - 1/2$

$\Rightarrow -t/RC = \ln(0.5)$

$\Rightarrow t = \overset{\cancel{0.6}\; 7\,20}{0.6} \times 300 \times 50 \times 10^{-6}$

$= 9 \times 10^{-3}$

$q = q_0 (1 - e^{-t/RC})$

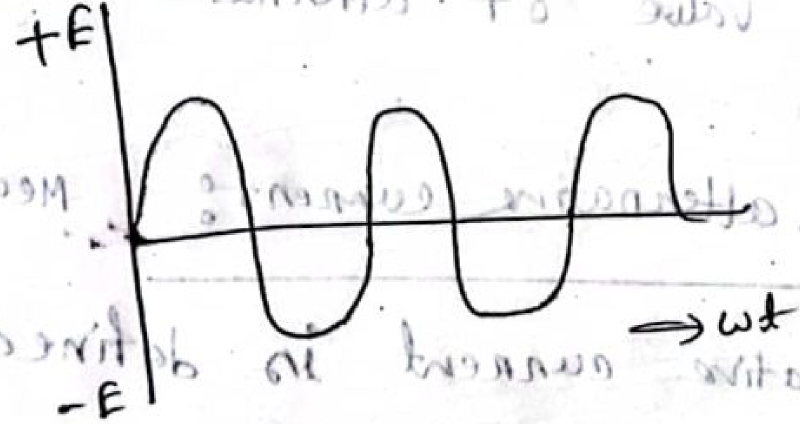
$0.909 q_0 = q_0 (1 - e^{-t/RC})$

$= \ln(1 - 0.90)$

$2.3 \times 300 \times 50 \times 10^{-6}$

0.0345

ng. current:

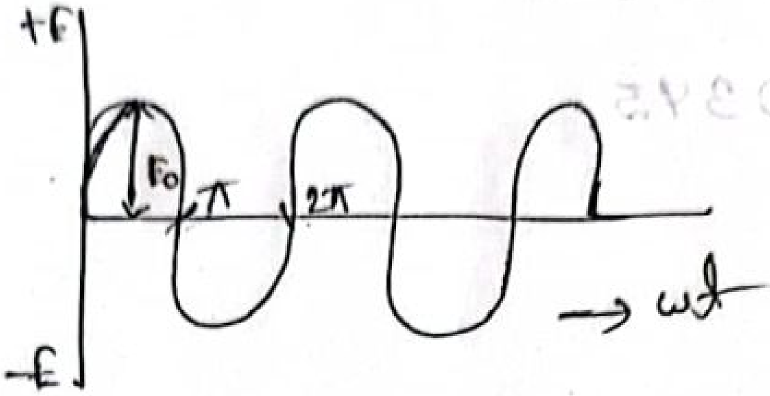


$I_0 \sin\omega t$

$E_0 \sin\omega t$

va

Peak value of alternating current:



The maximum value of alternating current or emf in the positive or negative direction is called peak value of alternative current.

Mean value of alternative current: Mean value of alternative current is defined as its average over half a cycle.

$$I_{mean} = \frac{\int_0^{T/2} I\,dt}{T/2}$$

Here,

$$T = \frac{2\pi}{\omega}$$

$$T/2 = \frac{\pi}{\omega}$$

$$I_{mean} = \int_0^{\pi/\omega} I_0 \sin \omega t$$

$$= \frac{I_0 \omega}{\pi} \int_0^{\pi/\omega} \sin \omega t \, dt$$

$$= \frac{I_0 \omega}{\pi} \int_0^{\pi/\omega} \frac{-\cos \omega t}{\omega} \, . dt$$

$$= \frac{I_0 \omega}{\pi} \left[\frac{-\cos \omega t}{\omega} \right]_0^{\pi/\omega}$$

$$\cancel{= \frac{I_0 \omega}{\pi} \left[\frac{\cos \frac{\pi}{\omega} t}{} - \cos 0 \right]}$$

$$= \frac{I_0 \omega}{\pi} \left[-\cos \omega \frac{\pi}{\omega} + \cos \omega \times 0 \right]$$

$$= \frac{I_0}{\pi} [1 + 1]$$

$$= \frac{2 I_0}{\pi}$$

Root mean square value of alternative current :-

It is define as the square root of the average of I^2 during a complete cycle.

$$\overline{I}^2_{mea} = \frac{\int_0^{2\pi/\omega} I^2\, dt}{2\pi/\omega}$$

$$= \frac{\int_0^T I^2 dt}{T}$$

$$= \frac{\int_0^{2\pi/\omega} I_0^2 \sin^2\omega t\, dt}{2\pi/\omega}$$

$$= \frac{I_0^2\omega}{2\pi}\int_0^{2\pi/\omega} \sin^2\omega t\, dt$$

$$= \frac{I_0^2\omega}{2\pi}\, 1/2 \int_0^{2\pi/\omega} 2\sin^2\omega t\, dt$$

$$= \frac{I_0^2 \omega}{4\pi} \int_0^{2\pi/\omega} (1 + \cos 2\omega t)\, dt$$

$$= \frac{I_0^2 \omega}{4\pi} \left[\int_0^{2\pi/\omega} dt - \int_0^{2\pi/\omega} \cos 2\omega t \, dt \right]$$

$$= \frac{I_0^2 \omega}{4\pi} \left[[t]_0^{2\pi/\omega} - \left[\frac{\sin 2\omega t}{2\omega} \right]_0^{2\pi/\omega} \right)$$

$$= \frac{I_0^2 \omega}{4\pi} \left[\left(\frac{2\pi}{\omega} - 0 \right) - \left(\frac{\sin 2\omega \cdot \frac{2\pi}{\omega}}{2\omega} - \frac{\sin 2\omega \cdot 0}{2\omega} \right) \right]$$

$$= \frac{I_0^2 \omega}{4\pi} \left[\frac{2\pi}{\omega} - \left(\frac{\sin 2\pi}{\omega} - 0 \right) \right.$$

$$\cancel{I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707\, I_0}$$

$$= \frac{I_0^2 \omega}{4\pi} \cdot \frac{2\pi}{\omega}$$

$$I^2 = \frac{I_0^2}{2}$$

$$I = \sqrt{\frac{I_0^2}{2}}$$

$$= I_0 / \sqrt{2}$$

From factor,

$$\therefore I_{rms} = 0.707\, I_0$$

$$= \frac{I_{rms}}{I_{mean}} = \frac{0.707\, I_0}{0.637} = 1.11$$

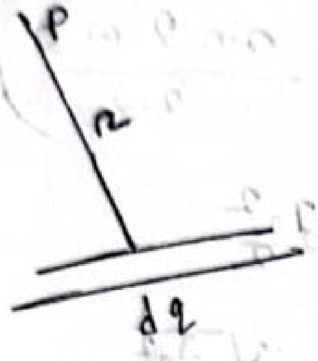
(Q) Show that the from factor is 1.11.

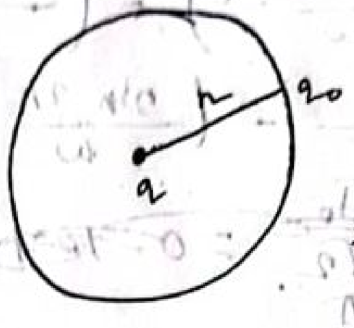
[I_{mean}, I_{rms} ব্যবহার করে]

(4)

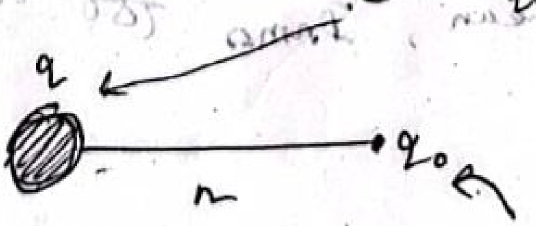
Change and electric potential

Electric ~~field~~ field / Electric field strength:





$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2}$$



Electric field: Electric field is defined as the force per unit change.

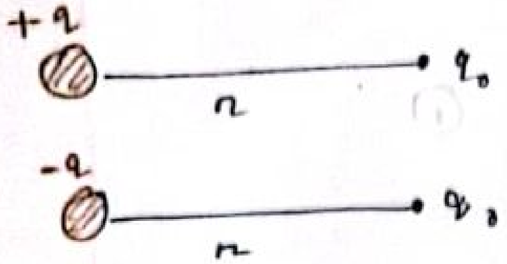
[point change + ...]

$$E = \frac{F}{q_0}$$

[unique change ... + ...]

$$E = \frac{F}{q_0}$$

Electric field due to a point charge.

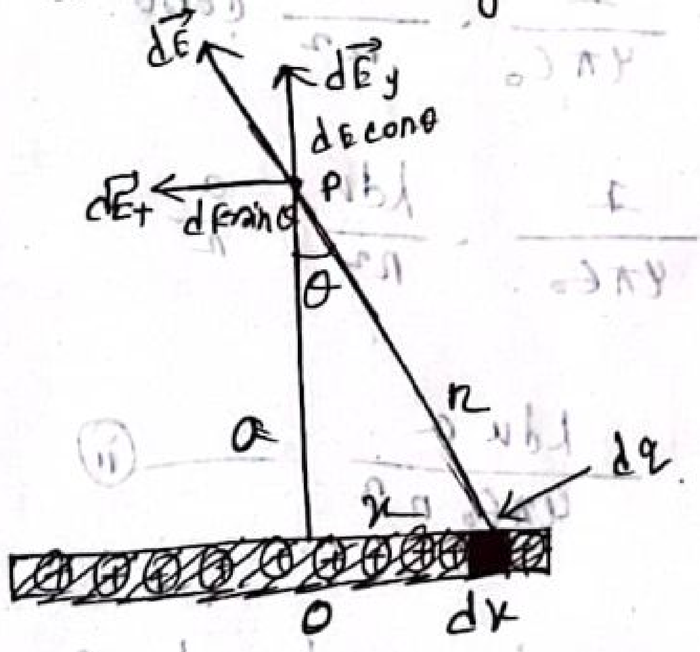


$$E = \frac{F}{q_0}$$

$$= \frac{\frac{1}{4\pi\epsilon_0} \cdot \frac{q q_0}{r^2}}{q_0}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}.$$

Electric field due to a long uniformed charge wire.



Let us, consider long uniformed charge wire having charge, λ too long per meter

So, the magnitude of the electric field at P. due to charge $dq = \lambda\, dx$.

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda\, dx}{r^2} \quad \text{--- (i)}$$

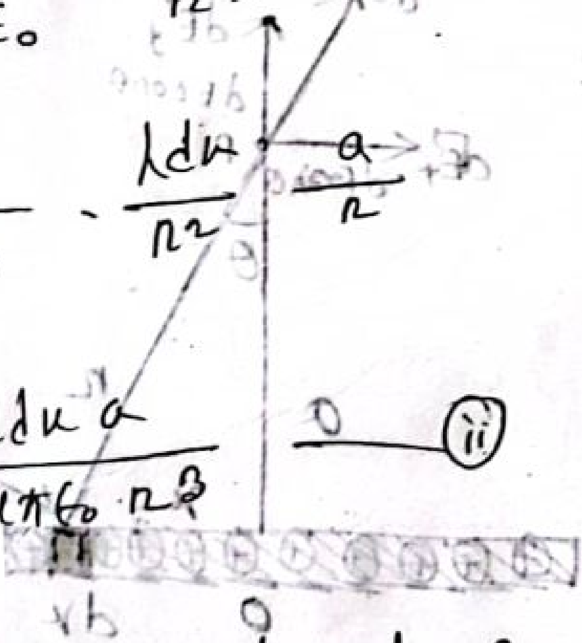
Here,

Only the y component can contribute to the resultant field at P.

$$d\vec{E} \cos\theta = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda\, dx}{r^2} \cos\theta$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda\, dx}{r^2} \cdot \frac{a}{r}$$

$$= \frac{\lambda\, dx\, a}{4\pi\epsilon_0\, r^3} \quad \text{--- (ii)}$$



Therefore the electric field at P.

$$E_P = \int_{-\alpha}^{+\alpha} dE \cos\theta$$

$$= \int_{-\alpha}^{+\alpha} \frac{\lambda\, dx\, a}{4\pi\epsilon_0 r^3}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\alpha}^{+\alpha} \frac{dx}{r^3}$$

$$\boxed{\begin{aligned} r^2 &= a^2 + x^2 \\ r^3 &= (a^2 + x^2)^{3/2} \end{aligned}}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\alpha}^{+\alpha} \frac{dx}{(a^2 + x^2)^{3/2}} \quad \text{(ii)}$$

Let,

$$x = a \tan\theta$$

$$dx = a \sec^2\theta \, d\theta$$

When

$$x = -\alpha, \ \theta = -\frac{\pi}{2}$$

$$x = \alpha, \ \theta = \frac{\pi}{2}$$

$$E_p = \frac{\lambda a}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{a \sec^2\theta \, d\theta}{(a^2 + a^2 \tan^2\theta)^{3/2}}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{a \sec^2\theta}{\{a^2(1 + \tan^2\theta)\}^{3/2}}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{a \sec^2\theta \, d\theta}{(a^3 \sec^3\theta)\, d\theta}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{d\theta}{a^2 \sec\theta}$$

$$= \frac{\lambda a}{4\pi\epsilon_0 . a^2} \int_{-\pi/2}^{\pi/2} \frac{d\theta}{\sec\theta}$$

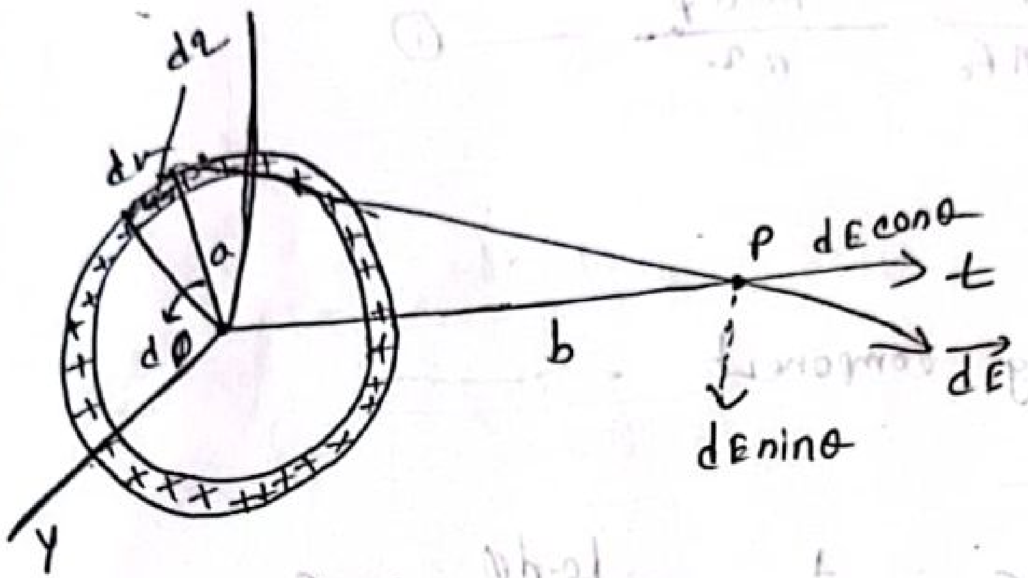
$$= \frac{\lambda \cancel{a}}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \cos\theta \, d\theta$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left[\sin\theta\right]_{-\pi/2}^{\pi/2}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left[\sin\frac{\pi}{2} + \sin\frac{\pi}{2}\right]$$

$$= \frac{\lambda}{4\pi\epsilon_0} \times 2 = \frac{\lambda}{2\pi\epsilon_0 a}$$ Ans

Electric field at a point on the axis of a charge circular form of wire.



Let us, consider circular form of a wire of radius a charge with λ coloumb per meter.

$$dQ = \lambda \times dr$$

$$= \lambda \times (r \times \theta)$$

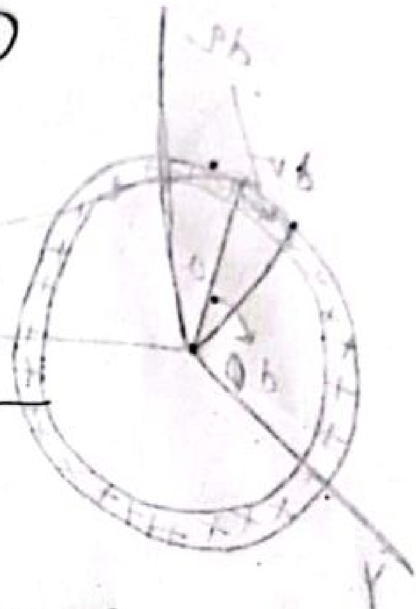
$$= \lambda \times a \times d\phi$$

$$\therefore d\varphi = \lambda a \, d\varphi$$

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\lambda a\, d\phi}{r^2} \quad \text{--- (i)}$$

Here,



Only the y component

$$dE \cos\theta = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda a\, d\phi}{r^2} \cdot \cos\theta$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda a\, d\phi}{r^2} \frac{b}{r}$$

$$= \frac{\lambda a b}{4\pi\epsilon_0 r^3} d\phi \quad \text{--- (ii)}$$

Therefore, the elect ~ P,

$$E_P = \int_{-\alpha}^{+\alpha} dE \cos\theta$$

$$= \int_0^{2\pi} \frac{\lambda ab}{4\pi\epsilon_0 r^3} \, d\phi$$

$$= \frac{\lambda ab}{4\pi\epsilon_0} \int_0^{2\pi} \frac{d\phi}{r^3}$$

$$= \frac{\lambda ab}{4\pi\epsilon_0} \int_0^{2\pi} \frac{d\phi}{(a^2+b^2)^{3/2}}$$

$$= \frac{\lambda ab}{4\pi\epsilon_0 (a^2+b^2)^{3/2}} \int_0^{2\pi} d\phi$$

$$= \frac{\lambda ab}{4\pi\epsilon_0 (a^2+b^2)^{3/2}} \left[\phi\right]_0^{2\pi}$$

$$= \frac{\lambda ab}{4\pi\epsilon_0 (a^2+b^2)^{3/2}} [2\pi - 0]$$

$$= \frac{\lambda a b}{2\epsilon_0 (a^2+b^2)^{3/2}}$$

If the total charge on the ring is

$q = 2\pi a \lambda$ then,

$$E_P = \frac{qb}{4\pi\epsilon_0 (a^2+b^2)^{3/2}} \quad \text{——— (4)}$$

if $b >> a$, then neglecting a^2 compared with b^2

$$E_P = \frac{q}{4\pi\epsilon_0 b^2} \quad \text{——— (5)}$$

This is the desired expression.

—o—

(ii) Electric Potential:

The electric potential at a point in an electric field is defined as the work done in bringing a unit positive charge from infinity to that point in ques. If w be the work done in bringing a positive charge from infinity to the point, then the electric potential,

$$V = \frac{W}{q_0} = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

Electric potential energy:

$$U = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r}$$

For +q charge,

Electric potential, $V_1 = \frac{1}{4\pi\epsilon_0}\frac{+q}{r_1}$

For −q charge,

$$V_2 = \frac{1}{4\pi\epsilon_0}\frac{-q}{r_2}$$

The next electric potential V due to is given by-

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r_1} - \frac{q}{r_2}\right) \quad \text{—— (i)}$$

If $r >> l$ then r_1 and r_2 may be written as

$$r_1 = r - l\cos\theta$$

$$r_2 = r + l\cos\theta$$

Substituting these values in eqn (i) we get,

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r - l\cos\theta} - \frac{q}{r + l\cos\theta}\right)$$

$$= \frac{1}{4\pi\epsilon_0}\left(\frac{q \cdot 2l\cos\theta}{r^2 - l^2\cos\theta}\right)$$

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$$= \frac{1}{4\pi\epsilon_0} \frac{2ql\cos\theta}{r^2}$$

$$p = q\, d$$
$$= q\, 2l$$

$$= \frac{1}{4\pi\epsilon_0} \frac{p\cos\theta}{r^2} \quad \text{---} \text{(ii)}$$

The electric field to p due to the dipole may now be evaluate. It is convenient to use polar co-ordinates (r, θ) in place of cartesian co-ordinates (x, y) for evaluating the components E_r and E_θ of the electric field are given by,

$$E_r = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r}\left(\frac{1}{4\pi\epsilon_0} \frac{p\cos\theta}{r^2}\right)$$

$$= -\frac{1}{4\pi\epsilon_0} p\cos\theta \cdot \frac{d}{dr}\left(\frac{1}{r^2}\right)$$

$$= -\frac{1}{4\pi\epsilon_0} p\cos\theta \cdot (-2) r^{-3}$$

$$= \frac{1}{4\pi\epsilon_0} p\cos\theta \frac{2}{r^3}$$

$$= \frac{2p\cos\theta}{4\pi\epsilon_0 r^3} \quad \text{---} \text{(iii)}$$

$$E_\theta = -\frac{1}{r}\frac{dV}{d\theta}$$

$$= -\frac{1}{r}\frac{d}{d\theta}\left(\frac{1}{4\pi\epsilon_0}\frac{p\cos\theta}{r^2}\right)$$

$$= -\frac{1}{r}\cdot\frac{1}{4\pi\epsilon_0}\frac{p}{r^2}\left[\frac{d}{d\theta}(\cos\theta)\right]$$

$$= -\frac{p}{4\pi\epsilon_0 r^3}(-\sin\theta)$$

$$= \frac{1}{4\pi\epsilon_0}\cdot\frac{p\sin\theta}{r^3} \quad \text{—— (iv)}$$

$$E = \sqrt{(E_r)^2 + (E_\theta)^2}$$

$$= \frac{1}{4\pi\epsilon_0}\sqrt{\left(\frac{2p\cos\theta}{r^3}\right)^2 + \left(\frac{p\sin\theta}{r^3}\right)^2}$$

$$= \frac{1}{4\pi\epsilon_0}\sqrt{\frac{4p^2\cos^2\theta}{r^6} + \frac{p^2\sin^2\theta}{r^6}}$$

$$= \frac{1}{4\pi\epsilon_0}\sqrt{\frac{4p^2\cos^2\theta + p^2\sin^2\theta}{r^6}}$$

$$= \frac{1}{4\pi\epsilon_0}\frac{p}{r^3}\sqrt{4\cos^2\theta + \sin^2\theta}$$

$$\frac{1}{4\pi\epsilon_0}\frac{p}{r^3}\sqrt{4\cos^2\theta+1-\cos^2\theta}$$

$$=\frac{1}{4\pi\epsilon_0}\frac{p}{r^3}\sqrt{1+3\cos^2\theta} \quad \text{——} \quad (v)$$

The angle

$$\alpha=\tan^{-1}\left(\frac{E_\theta}{E_r}\right).$$

$$=\tan^{-1}$$

(6) A diople is consisting of an electron and proton, 4×10^{-10} m apart. compute the electric field at a distance of 2×10^{-8} m on a line making perpendicular with the diople axis from the centre of the diople.

Answer:

~~Distance, d = 2 × 10^-8 m~~

Electric field,

$$E = \sqrt{(E_r)^2 + (E_\theta)^2}$$

$$= \frac{1}{4\pi\epsilon_0}\sqrt{\left(\frac{2p\cos\theta}{r^3}\right)^2 + \left(\frac{p\sin\theta}{r^3}\right)^2}$$

$$= \frac{1}{4\pi\epsilon_0}\sqrt{\frac{4p^2\cos^2\theta}{r^6} + \frac{p^2\sin^2\theta}{r^6}}$$

$$= \frac{1}{4\pi\epsilon_0}\cdot\frac{p}{r^3}\sqrt{4\cos^2\theta + \sin^2\theta}$$

$$= \frac{1}{4\pi\epsilon_0}\cdot\frac{p}{r^3}\sqrt{4\cos^2\theta + (1-\cos^2\theta)}$$

$$= \frac{p}{4\pi\epsilon_0 r^3}\sqrt{1 + 3\cos^2\theta}$$

$$= \frac{1}{4\pi\epsilon_0}\cdot\frac{2p}{r^3}$$

$$= \frac{1}{4\pi\epsilon_0}\cdot\frac{4\times10^{-10}\times1.6\times10^{-19}}{(2\times10^{-8})^3}$$

$$= 72000$$

Ex: 1 A diople is consisting of an electron and proton 4×10^{-10} m apart. compute the electric field at a distance of 2×10^{-8} m on a line making an angle of $45°$ with the dipole axis from the centre of the dipo.

Ans Dipole moment, $p = q \times 2d$

$$= (1.6 \times 10^{-19}) \times (4 \times 10^{-10})$$

$$= 6.4 \times 10^{-29} \text{ cm.}$$

Electric field, $E = \frac{p}{4\pi\varepsilon_0 r^3}\sqrt{1 + 3\cos^2\theta}$

$$= \frac{(9\times10^{9})\times(6.4\times10^{-29})}{(2\times10^{-8})^3}\sqrt{1+\cos^2 45}$$

$$= 1.138 \times 10^{5} \text{ NC}^{-1}$$

23). calculate the potential and field due to a diople moment 2×10^{-27} coul-meter at a point distance 2 cm from it (i) on its axis ii) on its perpendicular bisector.

Ans Electric potential,

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

=

Ans, $\vec{P} = 2 \times 10^{-27}$

$r = 2\text{ cm} = 0.02\text{ m}$.

$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$

(i) on it's axis,

$\theta = 0°$.

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{p\cos\theta}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{2 \times 10^{-27} \times \cos 0°}{(0.02)^2}$$

$= 4.5 \times 10^{-14}$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^3}\sqrt{1+3\cos^2\theta}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{2\times10^{-27}}{(0.02)^3}\sqrt{1+3(\cos 0)^2}$$

$$= 4.5\times10^{-12}$$

(ii) On it's perpendicular bisector, $\theta = 90^\circ$

$$V = \frac{1}{4\pi\epsilon_0} \frac{p\cos 90^\circ}{(0.02)^2}$$

$$= 0$$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^3}\sqrt{1+3\cos 90^\circ}$$

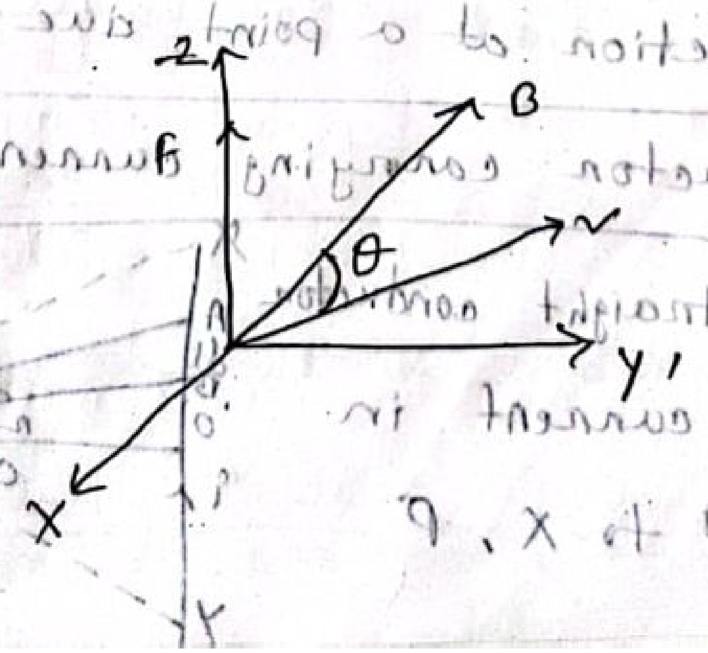
$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{2\times10^{-27}}{(0.02)^3} \cdot 1$$

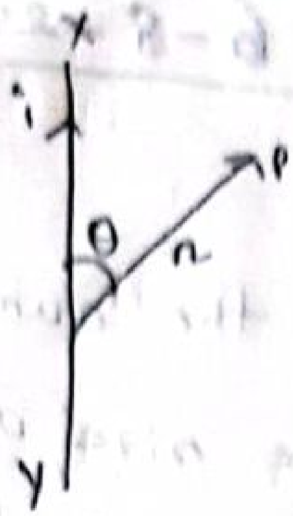
$$= 2.25\times10^{-12}$$

6-8-23

Magnetic field: The space around the current carrying conductor is defined as the size of a magnetic field. Magnetic field is a vector field whose magnitude and direction at any point are specified by a vector B called magnetic induction.

The definitation of B: consider a charge q moving with velocity v in a region where B exists. The charged particle experiences a force F in a direction perpendicular to both v and B.





$$dB \propto i$$

$$dB \propto \frac{1}{r^2}$$

$$dB \propto dl$$

$$dB \propto \sin\theta$$

$$\therefore dB \propto \frac{i\,dl\sin\theta}{r^2}$$

$$\Rightarrow dB = k\,\frac{i\,dl\sin\theta}{r^2}$$

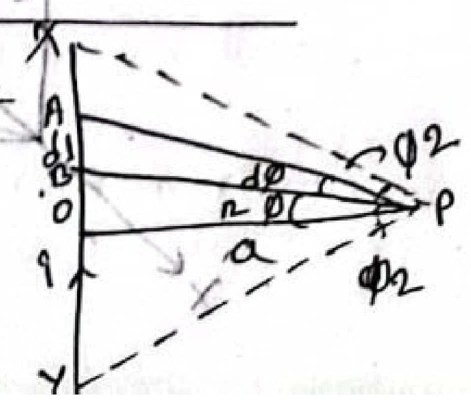
Where k is a constant, The value of k is $\frac{\mu_0}{4\pi}$. The total magnetic induction $\vec{B}$ at P due to the current carrying in entire element of the conductor is

$$\vec{B} = \frac{\mu_0}{4\pi}\, i \int \frac{\vec{dl} \times \vec{r}}{r^3}$$

Application of Biot-savart law:

Magnetic induction at a point due to a straight conductor carrying current

Consider a straight conductor XY carrying a current in the direction Y to X, P



is a point at a perpendicular distance a from the conductor. consider an element AB of length dl, Let $B = r$ and,

$\angle OBP = \theta$

Magnetic induction at P due to the element AB may

$$= dB$$

$$= \left(\frac{\mu_0}{4\pi}\right) \frac{i\,dl \sin\theta}{r^2} \quad \text{--- (1)}$$

From B, draw BC perpendicular to PA.

Let, $\angle OPB = \phi$, $\angle BPA = d\phi$

Then, $BC = dl \sin\theta = r\,d\phi$

$\triangle ABC$,

$$\sin\theta = \frac{BC}{dl} \Rightarrow BC = dl \sin\theta$$

$$\sin d\phi = \frac{BC}{BP} \Rightarrow BC = r\,d\phi$$

putting the value of $dl \sin\theta = r\,d\phi$ in eqn (1);

$$dB = \left(\frac{\mu_0}{4\pi}\right) \frac{i\,r\,d\phi}{r^2} = \left(\frac{\mu_0}{4\pi}\right) \frac{i\,d\phi}{r}$$

In $\triangle OPB$, $\cos\phi = a/r$ or $r = a/\cos\phi$

$$\therefore dB = \left(\frac{\mu_0}{4\pi}\right)\frac{i\cos\phi\, d\phi}{a} \quad \text{(ii)}$$

The direction of dB will be perpendicular to the plane containing dl and r. It will be directed into the page at P as shown by right hand rule. Let ϕ_1 and ϕ_2 be the angles made by the ends of the wire at P. Then magnetic reduction at P due to the whole conductor is -

Next page

is a point, at a perpendicular distance a from the conductor. Consider an element AB of length dl. Let BP = r and

$$B = \int_{-\phi_2}^{\phi_2} \left(\frac{\mu_0}{4\pi}\right) \frac{i\cos\phi \, d\phi}{a} = \left(\frac{\mu_0}{4\pi}\right) \frac{i}{a} [\sin\phi]_{\phi_1}^{\phi_2}$$

$$= \left(\frac{\mu_0}{4\pi}\right) \frac{i}{a} [\sin\phi_2 - \sin(-\phi_1)]$$

$$= \left(\frac{\mu_0}{4\pi}\right) \frac{i}{a} [\sin\phi_2 + \sin\phi_1)$$

If the conductor is infinity long $\phi_1 = \phi_2 \approx 90°$

$$\therefore B = \left(\frac{\mu_0}{4\pi}\right) \frac{i}{a} [\sin 90° + \sin 90°]$$

$$= \frac{\mu_0}{4\pi} \frac{2i}{a}$$

$$B' = \left(\frac{\mu_0}{4\pi}\right) \frac{i}{a} [\sin\phi_2 + \sin\phi_1]$$

$$= \left(\frac{\mu_0}{4\pi}\right) \left(\frac{i}{d/2}\right) [\sin 45 + \sin 45]$$

$$= \left(\frac{\mu_0}{4\pi}\right) \frac{i}{(d/2)} \times \frac{2}{\sqrt{2}}$$

$$= 4 \frac{\mu_0}{\pi} \cdot \frac{i}{\sqrt{2}d}$$

$$= \frac{\mu_0 i}{\sqrt{2}\pi d}$$

$$\therefore B = 4B' = \frac{2\sqrt{2}\mu_0 i}{\pi d}$$

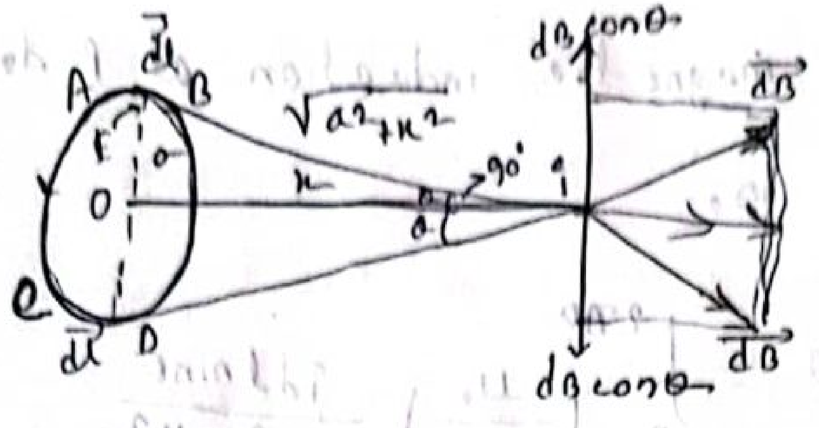
(d) $B = \frac{2\sqrt{2}\mu_0 i}{\pi . d} = \frac{2\sqrt{2} \times (4\pi \times 10^{-7}) \times 1}{\pi \times 1}$

$$= 8\sqrt{2} \times 10^{-7}\ Wb\,m^{-2}$$

10.4

Magnetic Induction at a point on the axis of a circular coil carrying current:

Consider a circular coil of radius a, carrying a current. P is a point on its axis at a distance x from the center O. Consider two opposite current elements AB and CD each of length dl. The distance of P from any point on the circumference of the coil is $\sqrt{a^2+x^2}$



field at P due to $AB = dB = \left(\frac{\mu_0}{4\pi}\right)\frac{i\,dl}{(a^2+x^2)}$

he direction of the current is ——— ①

right angles to the line joining P to AB.)

in the direction PL, perpendicular to the

joining the midpoint of AB with P.

sidering the element CD, the magnitude of

at P due to this element is the same

that given eqn (1). But it is directed

ng PM.

is the mid point of AB. Let $\angle EPO = \theta$,

he components $dB\cos\theta$ perpendicular to the

is of the coil and due to the two opposite

ments cancel each other. But components $dB\sin\theta$

long the axis are in the same direction.

Thus, the total magnetic induction at P due the entire coil is,

$$B = \int_0^{2\pi a} dB\sin\theta = \int_0^{2\pi a} \left(\frac{\mu_0}{4\pi}\right) \cdot \frac{i\,dl\sin\alpha}{a^2+x^2}$$

$$= \int_0^{2\pi a} \left(\frac{\mu_0}{4\pi}\right) \frac{i\,dl}{a^2+x^2} \times \frac{a}{(a^2+x^2)^{1/2}}$$

$$= \left(\frac{\mu_0}{4\pi}\right) \frac{i\,dl}{(a^2+x^2)^{3/2}} \times 2\pi a$$

$$= \frac{\mu_0 i a^2}{2(a^2+x^2)^{3/2}}$$

If the coil has N turns, then,

$$B = \frac{\mu_0 N i a^2}{2(a^2+x^2)^{3/2}}$$

At the center of the coil, $x = 0$

$$\therefore B = \frac{\mu_0 N i}{2a}$$

When,

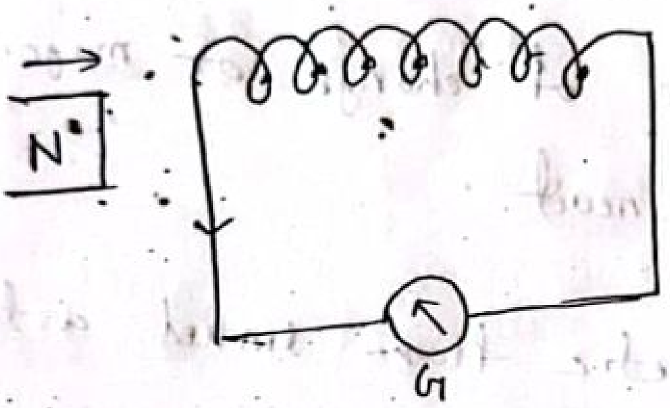
$x \gg a$ $$B = \frac{\mu_0 N i a^2}{2x^3}$$

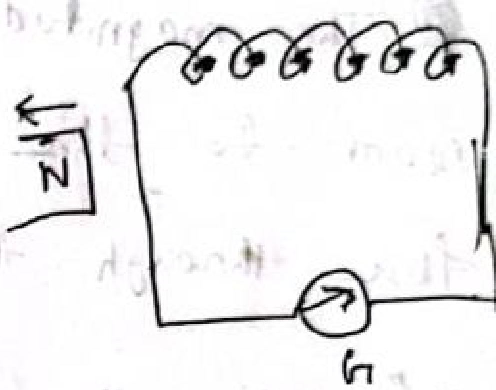
13-08-23

~~(*) Electro-magnetic Induction :~~

13-08-23

(*) Electromagnetic Induction:





Whenever magnetic lines of force are cut by a closed circuit, an induced current flows in t circuit. The current last outs so long as the flow is changing. The emf which produce this current is called induced emf. This phanemenon is called electromagnetic induction.

Panadane law of electromagnetic induction:

① Wheneven the magnetic flux through a conducton is change and emf is induced in the conducton.

② The magnitude of the induced emf is equal to the rate of change of magnetic flux through the circuit.

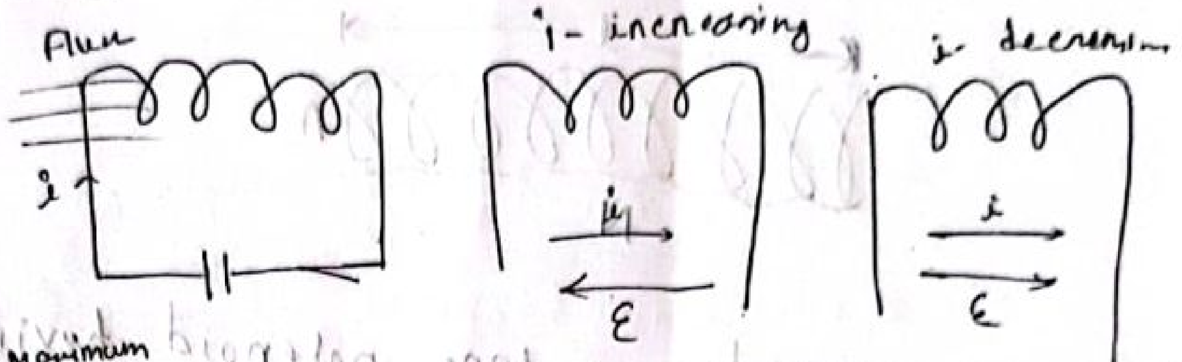
If ϕ is the magnetic flux linked with the circuit at any instant t and ε is the induced emf.

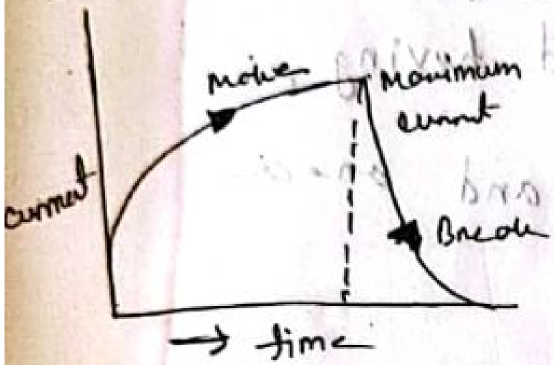
② The direction of the induced emf on current is such as to oppose the change that produce it. This is also called lenz's law. combine the both law $\varepsilon = \frac{d\phi}{dt}$

CamScanner

14-08-19

Self Induction:





This phenomenon is called self - induction

Self - inductance:

$$\phi \propto I$$

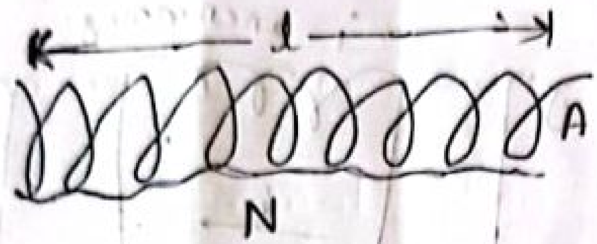
As $\phi = LI$ [L = Self Inductance]

$$\therefore L = \frac{\phi}{I} = \frac{\phi}{1\,amp}$$

$$\therefore L = \phi$$

[1 amp current এ
produce ~~[illegible]~~ হয়
[illegible] flux এর change হয়
~~[illegible]~~ self inductance]

Self inductance of a long solenoid:



Lets us consider a long solenoid having length l, number of turns N and area of construction A. current I

magnetic field / magnetic induction inside the solenoid,

$$\cancel{B = \mu_0 \left(\frac{N}{l}\right) I} \qquad B = \mu_0 \left(\frac{N}{l}\right) I$$

Flux through each turn $\phi = BA$

$$= \mu_0 \left(\frac{N}{l}\right) I \times A$$

through N turns $= \mu_0 \left(\frac{N^2}{l}\right) I A$

f inductance of the solenoid is,

$$L = \frac{\phi}{I} = \frac{\mu_0 N^2 I A}{l} \times \frac{1}{I}$$

$$= \frac{\mu_0 N^2 A}{l}$$

onstant permeability,

$$L = \frac{\mu N^2 A}{l} \text{; where, } \mu = \mu_0 \mu_r.$$

or

$$L = \frac{\mu_0 N^2}{l} \left[\mu_{r_1} A_1 + \mu_{r_2} A_2 + \mu_{r_3} A_3 + \cdots \right]$$

$$\left[\mu_0 = 4\pi \times 10^{-7} \right]$$

(7) Solenoid

$\mu_0 = 4\pi \times 10^{-7}$

length = 10cm

Number of turns = 100

Area of construction = 5 2 cm

Self inductance = ?

Solution.

∴ Self inductance,

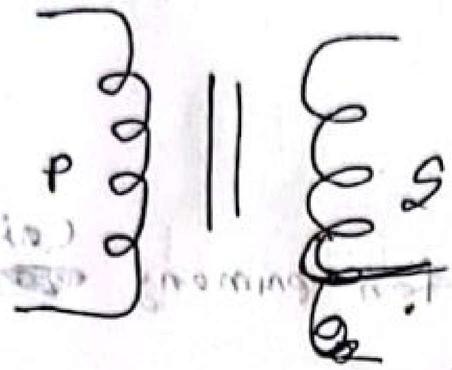
$$L = \frac{\mu_0 N^2 A}{l}$$

$$= \frac{4\pi \times 10^{-7} \times (100)^2 \times 25}{10}$$

$$= \frac{4\pi \times 10^{-7} \times (100)^2 \times 0.25}{0.1}$$

$$= 0.0314$$

(a) Mutual Induction:



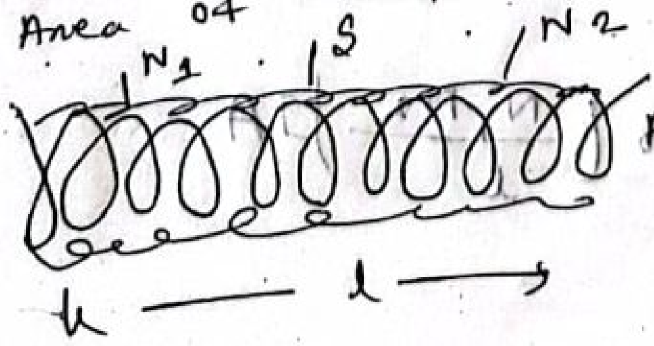
mutual inductance / co-efficient of mutual induction

$\phi \propto I$

$\phi = MI$

$M = \frac{\phi}{I}$ [M = Mutual Inductance]

Mutual inductance between two coaxial solenoid length l, number of turns of primary coil N_1, secondary coil N_2, Area of construction A, ~~ctot~~



Therefore the magnetic induction for solenoid,

$$B = \mu_0 \left(\frac{N}{l}\right) I$$

Flux through each turns for primary coil.

$$\phi = BA$$
$$= \mu_0 \left(\frac{N_1}{l}\right) I A$$

Flux through each turns for secondary coil,

$$\phi = BA$$
$$= \mu_0 \left(\frac{N_1}{l}\right) I A$$

Total flux through N_2 turns for secondary coil,

$$\phi = \mu_0 \left(\frac{N_1 N_2}{l}\right) I A$$

Mutual Induction,

$$M\Phi = \frac{\mu_0 (N_1 N_2) A}{l}$$

$$\Rightarrow M = \frac{\mu_0 N_1 N_2 A}{l}$$

$$\Rightarrow M = \frac{\mu_0 \mu_r N_1 N_2 A}{l} \quad [\mu_1 A_{n_1} + \mu_0 A_{2} n_2 + \mu_0 A_3 n_3 + \dots]$$

(4) Given

length = 30 cm

· Area of construction = 10^2 cm

Number of turns, N_1 = 1000

constant permeability, μ_r = 600

Number of turns, N_2 = 500

mutual inductance

$$M = \frac{\mu_0 \mu_r N_1 N_2 A}{l}$$

$$= \frac{4\pi \times 10^{-7} \times \overset{600}{1000} \times 500 \times 1}{0.3}$$

$$= 1.25 \times 10^{10}.$$

(*) Sinonoid -

0.88

Number of turns = 1000

length = 1 m

Area of construction, A = 7 cm^2

$\mu_r = 100$

Self conductance

$A = 7\, cm^2$
$= 7 \times 10^{-4}$

$$L = \frac{\mu_0 \mu_r N^2 A}{l}$$

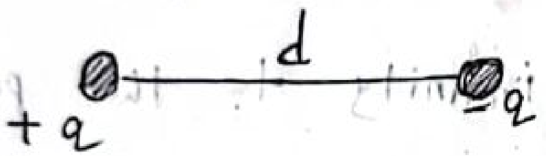
$$= \frac{4\pi \times 10^{-07} \times 100 \times (1000)^2 \times 7 \times 10}{1}$$

$= 0.088\, H$

Ex - 3 করতে হবে,

From - Books

(24) **Electric dipole:** When two equal and opposite charges placed at a small distance apart, then the arrangement is called as an electric dipole.



(25) **Electric dipole moment:**

$$P = qd$$

Electric potential and electric field due to an electric dipole: